Spectral Mobility and the Lorentz Factor from Projective Capacity Saturation

Jérôme Beau

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Abstract

The O-series saturation parameter η_n^{O7} is defined in [3] as the ratio of cumulative occupied projective capacity to residual frontier novelty. We show that this ratio is not an arbitrary fitting variable: under the Born–Infeld admissibility budget established in [2], a quadratic normalization of the two capacity components implies $\eta_n^{\mathrm{O7}} \to \beta\gamma$, where $\beta\gamma$ is the relativistic mobility and γ is the Lorentz factor. Consequently, the O-series profile $\Phi(\eta) = 1/\sqrt{1+\eta^2}$ converges to $1/\gamma$: the inverse Lorentz factor is the residual-capacity profile of the admissible spectral cascade. The quadratic normalization is not postulated; it is expected to follow from the Carnot–Carathéodory continuum geometry and co-metric closure established in [1, 4]. The main convergence lemma, identifying the O7 observables as the normalized components of the Born–Infeld budget, is stated precisely and left as the remaining technical task.

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1 O7 occupation and residual novelty

The O7 saturation parameter is defined in [3] by

$$\eta_n^{\mathrm{O7}}(x) = \frac{B_n(x)}{\Sigma_n(x)}, \qquad B_n(x) := |\psi_n(x)|^2 = \frac{|S_n \cap C_x|}{|C_x|}.$$

Here C_x is the effective resolution cell associated with the observable state x, and S_n is the BFS-reachable set up to cascade depth n.

The observable $B_n(x)$ is cumulative. It measures the fraction of the cell C_x already reached by the BFS cascade by depth n. In particular,

$$0 \le B_n(x) \le 1, \qquad B_{n+1}(x) \ge B_n(x).$$

It is therefore an occupation observable, not an infinitesimal mobility.

By contrast, $\Sigma_n(x)$ measures the average effective novelty contributed by the BFS frontier at depth n: the mean fraction of genuinely new projective direction brought by admissible paths reaching ∂S_n within C_x. It is a residual frontier observable, instantaneous and non-increasing in n.

The ratio η_n^{O7} is therefore not a fitting parameter. It is an occupied-over-residual capacity ratio:

$$\eta_n^{\mathrm{O7}} = \frac{\text{cumulative occupied capacity}}{\text{residual frontier novelty}}.$$

The question addressed in this note is whether this O7 ratio converges, in the continuum geometric limit, to the relativistic mobility ratio $dI_{\mathrm{space}}/dI_\tau$.

2 Algebraic Born–Infeld target

The cascade depth n is not a time coordinate. It is the ordinal depth of an admissible projective history, counting the number of distinguishable projective transitions undergone along that history [2].

Assume a bounded Born–Infeld mobility budget at each projective depth increment dn:

$$dI_\tau^2 + dI_{\mathrm{space}}^2 = c_{\mathrm{BI}}^2\, dn^2.$$

Define the normalized spectral mobility

$$\beta := \frac{dI_{\mathrm{space}}}{c_{\mathrm{BI}}\, dn}.$$

Then the residual temporal component is

$$dI_\tau = c_{\mathrm{BI}} \sqrt{1-\beta^2}\, dn,$$

and therefore

$$\frac{dI_{\text{space}}}{dI_\tau} = \frac{\beta}{\sqrt{1-\beta^2}} = \beta\gamma, \qquad \gamma := \frac{1}{\sqrt{1-\beta^2}}.$$

If a spectral saturation parameter η satisfies $\eta = dI_{\text{space}}/dI_\tau$, then

$$\Phi(\eta) = \frac{1}{\sqrt{1+\eta^2}} = \frac{1}{\gamma}.$$

The inverse Lorentz factor is the residual-capacity profile of a quadratic Born–Infeld budget. This step is purely algebraic. The non-trivial question is whether the O7 observables B_n and Σ_n converge to the two normalized components of this budget.

3 Why the reduced filling model is insufficient

The reduced filling model of [3] gives, before geometric correction,

$$B_n \simeq \frac{n}{D}, \qquad \Sigma_n \simeq 1 - \frac{n}{D},$$

equivalently $B_n + \Sigma_n \simeq 1$. This is a linear residual law. It describes monotone filling of a finite resolution cell but does not produce the Lorentzian residual factor. Indeed, setting $B_n \to \beta$ in the linear model would give $\Sigma_n \to 1-\beta$, whereas the Born–Infeld mobility budget requires $\Sigma_n \to \sqrt{1-\beta^2}$.

The linear filling model therefore captures the monotone occupation mechanism but not the continuum geometric correction responsible for the Lorentzian square root. The required structure is the quadratic residual law

$$B_n^2 + \Sigma_n^2 \to 1.$$

Under the identification $B_n \to \beta$, this gives $\Sigma_n \to \sqrt{1-\beta^2}$.

4 From linear filling to quadratic normalization via Q5b

4.1 Q5b input

The reduced filling model operates at the combinatorial level of a finite resolution cell. The Q5b continuum limit supplies the missing geometric structure [1].

In the pre-saturation large-q regime, BFS shells on $\mathrm{Heis}_3(\mathbb{Z}/q\mathbb{Z})$ converge to Carnot–Carathéodory balls on $\mathrm{Heis}_3(\mathbb{R})$. The linear residual law of the reduced model is a coarse filling approximation valid at finite D and finite q; it does not encode the sub-Riemannian anisotropy of the CC geometry.

The effective co-metric extracted from the completed principal-symbol geometry of the admissibility operator is Lorentzian in the homogeneous regime. After the Q11 co-metric closure [4], the normalized co-metric is

$$g^{\mu\nu} = 2\,\eta^{\mu\nu}, \qquad \eta = \mathrm{diag}(-1, +1, +1, +1),$$

with no free parameter. This Lorentzian structure imposes a quadratic normalization on the effective admissible components in the continuum limit — replacing the linear complement of the reduced filling model.

The chain is therefore:

CC convergence (Q5b)	$\Rightarrow$	continuum spatial and temporal components exist,
principal symbol / Q11 closure	$\Rightarrow$	quadratic normalization of effective components,
observable identification (Lemma 1)	$\Rightarrow$	$B_n^2 + \Sigma_n^2 \to 1$.

The geometric structure is not postulated; it comes from Q5b and Q11. The remaining task is the third step: showing that the O7 observables B_n and Σ_n are precisely the normalized projections of the spatially occupied and temporally residual components of the continuum budget.

4.2 Main convergence lemma

Lemma 1 (O7 observables as normalized CC budget components)**.** *Let $B_n(x) := |\psi_n(x)|^2 = |S_n \cap C_x|/|C_x|$ be the O7 cumulative occupation observable, and let $\Sigma_n(x)$ be the O7 residual frontier novelty [3]. In the pre-saturation large-q regime, under the Q5b BFS-to-CC convergence and the Q11 homogeneous co-metric closure, these observables converge to the normalized spatial and temporal components of the Born–Infeld mobility budget:*

$$B_n \longrightarrow \beta, \qquad \Sigma_n \longrightarrow \sqrt{1-\beta^2},$$

where $\beta = v_{\mathrm{eff}}/c_{\mathrm{BI}}$ is the normalized projected spectral mobility. Equivalently,

$$B_n^2 + \Sigma_n^2 \longrightarrow 1.$$

The remaining work is the observable-identification step. Q5b provides the continuum quadratic geometry. O7 provides the occupied-over-residual observables. The lemma asserts that these two descriptions refer to the same normalized capacity components in the common large-q, pre-saturation regime.

5 Consequence for the Lorentz factor

Theorem 1 (Lorentz factor as residual projective capacity)**.** *Assume the Born–Infeld mobility budget of Section 2 and the convergence of Lemma 1. Then the O7 saturation parameter satisfies*

$$\eta_n^{\mathrm{O7}} = \frac{B_n}{\Sigma_n} \longrightarrow \frac{\beta}{\sqrt{1-\beta^2}} = \beta\gamma,$$

and therefore the O-series saturation profile satisfies

$$\Phi\big(\eta_n^{\mathrm{O7}}\big) = \frac{1}{\sqrt{1+\big(\eta_n^{\mathrm{O7}}\big)^2}} \longrightarrow \frac{1}{\gamma}.$$

The special-relativistic residual factor is thus recovered from the O7 occupied-over-residual capacity ratio, provided the Q5b continuum geometry upgrades the reduced linear filling law to the quadratic Born–Infeld residual law. The Lorentz factor is not added as an external kinematical postulate. It is the effective reading of residual temporal capacity in a bounded projective budget.

Remark 1 (Saturation pressure at $v \to c$)**.** *Under the identification $\eta_n^{\mathrm{O7}} \to \beta\gamma$, the O7 saturation pressure $P_n^{\mathrm{sat}} = \gamma\,\eta_n$ satisfies $P_n^{\mathrm{sat}} \to \beta\gamma^2$, which diverges as $v \to c$. This is consistent with the light cone reading $dI_\tau = 0 \Leftrightarrow v = c$: the budget is fully consumed by spatial mobility and no temporal residual remains.*

Remark 2 (Scope)**.** *Theorem 1 directly yields time dilation ($d\tau/dn = 1/\gamma$) and the light cone ($dI_\tau = 0 \Leftrightarrow v = c$). Length contraction requires the additional reconstruction of effective Lorentz transformations between inertial observers. The extension to general relativity follows from the non-uniform regime: when localised configurations modify the local spectral budget, the position-dependent metric $g_{\mu\nu}(x)$ encodes gravitational time dilation as a comparison of residual temporal capacities along distinct projected histories.*

References

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